

JothirnadhReddy Peram | Mechatronics Engineering Student

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SUMMARY

A driven Mechatronics Engineering student (9.43 CGPA) operating at the intersection of robotics, embedded systems, and artificial intelligence. Passionate about integrating mechanical design, electronics, and software to develop autonomous and practical real-world solutions. Proven ability to ship full-stack hardware projects from CAD modeling to AI deployment. Driven by the motto: "Solve with Robots. Succeed with Business."

EDUCATION

SRM Institute of Science and Technology

Bachelor of Technology in Mechatronics Engineering, CGPA: 9.43

Chennai

Aug 2024 – May 2028

Sri Chaitanya College of Education

Intermediate Education, Grade: 92.2%

Andhra Pradesh

Jun 2022 – Mar 2024

TECHNICAL SKILLS

- **Programming:** Python, C
- **AI & Vision:** YOLOv11s, OpenCV, Generative AI
- **Robotics & IoT:** ESP32, Arduino, Sensor Fusion (IMU), PID Control, WebSockets, Blynk Platform
- **CAD & Hardware:** SOLIDWORKS, Pneumatics, Basic PCB Concepts, Motor Drivers

KEY PROJECTS

Semi-Autonomous Weed Detection & Laser Removal Robot

Feb 2026 – Mar 2026

Developed a precision agriculture robot combining AI inference, embedded control, and mechanical actuation. Implemented YOLOv11s and OpenCV for weed detection, extracting pixel coordinates from bounding boxes. Engineered a galvo-inspired servo laser targeting system by applying calibration-based coordinate transformations to map image targets to real-world angles for precise weed eradication.

AI Brain V1 (Multi-Agent Assistant)

Oct 2025

Built a modular, Jarvis-style multi-agent AI system utilizing the Mistral 7B Large Language Model. Integrated automated pipelines for email management, web browsing, weather/calendar tracking, and system monitoring.

Industrial Process Monitor

Apr 2026

Developed a local, cloud-independent industrial monitoring system using ESP32 microcontrollers. Enabled real-time tracking of pressure, tank levels, and motor RPM, streaming data directly to a browser-based dashboard via WebSockets.

Self-Balancing Robot

Nov 2024 – Dec 2024

Designed and built a two-wheeled self-balancing robot with a closed-loop PID control system in C++. Processed accelerometer and gyroscope data from an MPU6050 IMU for real-time vertical stability and tilt estimation.

Air-Powered Pneumatic Safety Bumper

Sep 2025 – Dec 2025

Modeled a complete mechanical assembly and pneumatic circuit system in SOLIDWORKS. Utilized double-acting cylinders to extend the bumper pre-impact, applying fluid mechanics principles for optimal kinetic energy absorption.

ACHIEVEMENTS & CERTIFICATIONS

- **Awards:** Awarded Best Hardware Design (Agricultural Robotics theme) at the National Level Automation & Robotics Challenge (ARC'26) organized by SRM Institute of Science and Technology.
- **Certifications:** Generative AI Masterminds (Outskill), 100 Days of Code: Complete Python Pro Bootcamp (Udemy)